FAITHFUL FLATNESS IN THE MIDDLE OF A COMPOSITE OF RING MAPS DOES NOT CHANGE TOR AMPLITUDE OVER THE BASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex over a ring, and its behaviour under derived tensor product. defined in the same section, source lines 17100-17236. Supplies the two quantities whose equality is asserted.
- A complex can be represented by a complex of A-modules and a module by a free resolution when computing derived tensor products. used at proof lines 17276-17279, where K is represented by a complex and M by a free resolution. Replaces the derived constructions by ordinary total complexes whose equality can be displayed.
- A module over a ring is zero if and only if its base change along a faithfully flat map is zero. used at the end of the proof, at proof lines 17280-17284, in the same way as in the preceding statement. Passes from the vanishing of the base changed cohomology modules to the vanishing of the cohomology modules over R.

Source reference. Original lemma 0DJF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16975–16975. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Given ring maps $R \to A \to B$ with $A \to B$ faithfully flat and $K \in D(A)$ the tor amplitude of K over R is the same as the tor amplitude of $K \otimes^{\mathbf{L}}_A B$ over R.*

Proof. This is true because for an R-module M we have $H^i(K \otimes^{\mathbf{L}}_R M) \otimes_A B = H^i((K \otimes^{\mathbf{L}}_A B) \otimes^{\mathbf{L}}_R M)$ for all i. Namely, represent K by a complex $K^\bullet$ of A-modules and choose a free resolution $F^\bullet \to M$. Then we have the equality

$$\mathrm{Tot}(K^\bullet \otimes_A B \otimes_R F^\bullet) = \mathrm{Tot}(K^\bullet \otimes_R F^\bullet) \otimes_A B$$

The cohomology groups of the left hand side are $H^i((K \otimes^{\mathbf{L}}_A B) \otimes^{\mathbf{L}}_R M)$ and on the right hand side we obtain $H^i(K \otimes^{\mathbf{L}}_R M) \otimes_A B$. □